BRENNAN'S CONJECTURE AND SHARP INVERSE-SQUARE INTEGRAL MEANS: A REVERSE-FORMALIZATION EXPOSITION OF THE `OAI.ANALYSIS.INTEGRALMEANS` LEAN 4 ARCHITECTURE

REVERSE-FORMALIZED FROM `OAI/ANALYSIS/INTEGRALMEANS` (LEAN 4 / MATHLIB)

ABSTRACT. We present a self-contained mathematical exposition of the proof of Brennan's Conjecture and the sharp inverse-square integral means identity $B(-2) = 1$ that follows the exact module DAG, definitions, lemmas, and theorem signatures of the 47-file, 14,759-line Lean 4 formalization in `OAI.Analysis.IntegralMeans`. Unlike a traditional human paper that appeals to high-level topological and analytic machinery—such as Montel's theorem, Hurwitz's theorem, Sard's theorem, the Jordan Curve Theorem, Stokes' theorem on punctured domains, and Lusin–Novikov measurable selection—the Lean 4 formalization introduces six systematic structural bypasses that reduce every step to constructive Mathlib primitives. We document each formal definition and theorem alongside its exact Lean 4 code and explain the formal bypasses in complete mathematical detail.

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1. Introduction and Formal Architecture

1.1. Statement of Main Theorems in Model.lean, Main.lean, SharpModel.lean, and SharpEndpoints.lean.

Let $\mathbb{D} = \{z \in \mathbb{C} : \|z\| < 1\}$ be the open unit disk. A function $f : \mathbb{C} \to \mathbb{C}$ is univalent on an open set $U \subseteq \mathbb{C}$ if it is complex differentiable on U and injective on U. The normalized schlicht class $\mathcal{S}$ consists of univalent functions on $\mathbb{D}$ satisfying $f(0) = 0$ and $f'(0) = 1$.

- **Core Analytic Model and Universal Spectrum** ($f, \varphi : \mathbb{C} \to \mathbb{C}$, $U, W \subseteq \mathbb{C}$, $t, r, p \in \mathbb{R}$):

```
def disk : Set ℂ := Metric.ball (0 : ℂ) 1

def UnivalentOn (f : ℂ → ℂ) (U : Set ℂ) : Prop :=
  DifferentiableOn ℂ f U ∧ InjOn f U

def Schlicht (f : ℂ → ℂ) : Prop :=
  UnivalentOn f disk ∧ f 0 = 0 ∧ deriv f 0 = 1

def integralMean (f : ℂ → ℂ) (t r : ℝ) : ℝ :=
  (2 * Real.pi)⁻¹ * ∫ θ in -Real.pi..Real.pi,
    ‖deriv f ((r : ℂ) * Complex.exp ((θ : ℂ) * Complex.I))‖ ^ t

def beta (f : ℂ → ℂ) (t : ℝ) : EReal :=
  Filter.limsup (fun r : ℝ =>
    ((Real.log (integralMean f t r) / Real.log (1 / (1 - r)) : ℝ) : EReal))
    (nhdsWithin 1 (Iio 1))

def spectrum (t : ℝ) : EReal :=
  sSup {b : EReal | ∃ f : ℂ → ℂ, Schlicht f ∧ beta f t = b}

def ConformalMapToDisk (W : Set ℂ) (φ : ℂ → ℂ) : Prop :=
  IsOpen W ∧ DifferentiableOn ℂ φ W ∧ BijOn φ W disk
```

Mathematically, for $f \in \mathcal{S}$, $t \in \mathbb{R}$, and $r \in (0,1)$, the radial integral mean, growth exponent, and universal integral means spectrum are defined in $\overline{\mathbb{R}}$ ('EReal') by

$$M_t(f,r) := \frac{1}{2\pi}\int_{-\pi}^{\pi} |f'(re^{i\theta})|^t \, d\theta, \qquad \beta_f(t) := \limsup_{r\to 1^-} \frac{\log M_t(f,r)}{\log \frac{1}{1-r}}, \qquad B(t) := \sup_{f\in\mathcal{S}} \beta_f(t).$$

- **Main Theorems in `Main.lean` and `SharpEndpoints.lean`** ($W \subseteq \mathbb{C}$, $\varphi : \mathbb{C} \to \mathbb{C}$, $p, t \in \mathbb{R}$):

```
theorem inverse_square_spectrum : spectrum (-2) = 1 :=
  inverse_spectrum_eq_one

theorem brennan_conjecture {W : Set ℂ} {φ : ℂ → ℂ}
    (hφ : ConformalMapToDisk W φ) {p : ℝ}
    (hp1 : 4 / 3 < p) (hp2 : p < 4) :
    IntegrableOn (fun z => ‖deriv φ z‖ ^ p) W volume :=
  conformal_domain_area hφ.1 hφ.2.1 hφ.2.2 hp1 hp2

theorem sharp_schlicht_area {t : ℝ} :
    (∀ f : ℂ → ℂ, Schlicht f →
      IntegrableOn (fun z => ‖deriv f z‖ ^ t) disk volume) ↔
      t ∈ Ioo (-2 : ℝ) (2 / 3)

theorem sharp_brennan_conjecture {p : ℝ} :
    (∀ W : Set ℂ, ∀ φ : ℂ → ℂ, ConformalMapToDisk W φ →
      IntegrableOn (fun z => ‖deriv φ z‖ ^ p) W volume) ↔
      p ∈ Ioo (4 / 3 : ℝ) 4
```

Mathematically:

(1) **Sharp Inverse-Square Spectrum**: $B(-2) = 1$.

(2) **Brennan's Conjecture**: For every open set $W \subseteq \mathbb{C}$ and conformal bijection $\varphi : W \to \mathbb{D}$, and every $p \in (4/3, 4)$,

$$\int_W |\varphi'(w)|^p \, dA(w) < \infty.$$

(3) **Sharpness of the Open Intervals**: Universal area integrability $\int_{\mathbb{D}} |f'(z)|^t \, dA(z) < \infty$ across $f \in \mathcal{S}$ holds if and only if $t \in (-2, 2/3)$, and universal domain integrability $\int_W |\varphi'(w)|^p \, dA(w) < \infty$ holds if and only if $p \in (4/3, 4)$, witnessed at the endpoints by the Koebe function $f_K(z) = z/(1-z)^2$ and its inverse $\varphi_K = f_K^{-1}$ on the slit plane $W_K = f_K(\mathbb{D}) = \mathbb{C} \setminus (-\infty, -1/4]$.

1.2. **Six Structural Bypasses in the Lean 4 Formalization.** A central reason the Lean 4 formalization in `OAI.Analysis.IntegralMeans` achieves a complete, self-contained proof in 14,759 lines is that it systematically replaces classical non-constructive or heavy geometric theorems with direct analytic and algebraic constructions tailored to Mathlib's library:

Bypass 1. No Montel or Hurwitz Theorem (`Distortion.lean`, `CompactClass.lean`): Instead of topologizing $\mathcal{K} \subset \mathrm{Hol}(\mathbb{H}, \mathbb{C})$ directly and invoking Montel's and Hurwitz's theorems, the formalization models the compact class on the unit disk as a subset `diskSchlichtSet` $\subset C(\mathbb{D}, \mathbb{C})$ equipped with the compact-open uniform structure (`UniformOnFun`). Compactness is proved directly from Mathlib's **Arzelà–Ascoli theorem** via explicit Koebe Lipschitz bounds ($|f'(z)| \le (1+R)/(1-R)^3$). Closedness of injectivity avoids Hurwitz's theorem entirely by proving a **quantitative uniform separation inequality** (`schlicht_uniform_separation`): for any $z \neq w \in \mathbb{D}$, applying Koebe's 1/4 theorem on the maximal disk centered at z yields an explicit positive lower bound $\|f(w) - f(z)\| \ge c(z,w) > 0$ valid uniformly across all $f \in \mathcal{S}$, which survives pointwise limits immediately.

Bypass 2. No Differentiating Horizontal Integrals Under the Integral Sign (`TraceIdentity.lean`): The human proof of Lemma 4.3 differentiates $h(y) = \int_{\mathbb{R}} g(x,y)\,dx$ twice with respect to y and integrates by parts. Formalizing parametric differentiation of improper integrals and boundary decay at $x = \pm\infty$ is tedious. Instead, `TraceIdentity.lean` introduces the explicit vector field (**trace flux**)

$$\mathbf{V}(x,y) = \big((y-\varepsilon)\partial_x g(x,y),\ (y-\varepsilon)\partial_y g(x,y) - g(x,y)\big),$$

whose divergence is identically $\operatorname{div}\mathbf{V} = (y-\varepsilon)\Delta g$. Applying Mathlib's **rectangular divergence theorem** (`integral_divergence_prod_Icc_of_hasFDerivAt_of_le`) on $[-R,R]\times[\varepsilon,R]$ directly relates the horizontal slice integral $\int_{-R}^{R} g(x,\varepsilon)\,dx$ to $\iint (y-\varepsilon)\Delta g\,dx\,dy$ in one stroke.

Bypass 3. No Pole Singularities in the Potential (`CriticalPoint.lean`, `CriticalCount.lean`): In the human paper, the potential $\ell(z) = \log V_\xi^F(z)$ has a logarithmic singularity at the unique preimage $z_0 = F^{-1}(\xi)$, requiring punctured domains and small circle excavations. In `CriticalCount.lean`, the formalization works instead with the globally C^2 **squared reciprocal potential**

$$u(z) := \frac{|F(z)-\xi|^2}{y^{2k}} \qquad (\texttt{squaredReciprocalPotential}),$$

and replaces $\nabla\ell$ by the globally smooth **regularized gradient** (`regularizedGradient`):

$$\mathcal{G}_{F,k,\xi}(z) := \frac{ik}{y}(F(z)-\xi)\overline{(G_F(z)-\xi)}.$$

On the strict sublevel $\{u < c\}$, $\mathcal{G}_{F,k,\xi}$ is everywhere smooth, coincides with a positive scalar multiple of $-\nabla u$ on $\{u > 0\}$, and has zeros at *both* the pole $F(z) = \xi$ (where its real Jacobian determinant is $+k^2y^{-2}|F'(z)|^2 > 0$, hence Poincaré index $+1$) and the true critical points $G_F(z) = \xi$ (where its real Jacobian determinant has sign opposite to $J_F(z)$).

Bypass 4. No Stokes' Theorem for the Poincaré–Hopf Degree Formula (`Degree.lean`): Rather than using differential forms and Stokes' theorem on a manifold with boundary to equate boundary winding numbers to interior zero indices, `Degree.lean` proves `windingNumber_finite_regular_zeros` on a convex set $K \subset \mathbb{C}$ by **inductive multiplicative zero surgery** (`removeZero` and `surgeryLog`). At each nondegenerate zero $a \in s$, it divides the vector field $X(z)$ locally by $z - a$ (if $\det_{\mathbb{R}} DX(a) > 0$) or $\overline{z-a}$ (if $\det_{\mathbb{R}} DX(a) < 0$), smoothly patches via a complex logarithm on an annulus (`surgeryLog`), and peels off one zero at a time until the modified field is nowhere vanishing on the convex set K—where its winding number is 0 by contractibility.

Bypass 5. No Jordan Curve Theorem (`Winding.lean`, `LevelIndex.lean`): To identify the sum of winding numbers of the oriented level-curve loops $\gamma_1,\dots,\gamma_m$ around z with the indicator $\mathbf{1}_{\{u(z)<c\}}$, a classical proof appeals to the Jordan Curve Theorem on each component of $\{u = c\}$. Instead, `LevelIndex.lean` defines the integer-valued discrepancy function on $\mathbb{C}\setminus\{u = c\}$:

$$\mathcal{J}(z) := \sum_{j=1}^{m} \operatorname{Ind}(\gamma_j, z) - \mathbf{1}_{\{z\in\mathbb{H}:u(z)<c\}},$$

and proves that across each regular level loop γ_j, both $\sum_i \operatorname{Ind}(\gamma_i, z)$ and $\mathbf{1}_{\{u<c\}}$ jump by $+1$ in the inward normal direction (`winding_jump_centred` and the maximum principle on `windingInterior`). Thus $\mathcal{J}$ extends to a **continuous function** $\mathbb{C}\to\mathbb{Z}$ on all of $\mathbb{C}$ that vanishes outside a large ball, and hence is identically 0 by connectedness of $\mathbb{C}$.

Bypass 6. No Lusin–Novikov Selection Theorem (`BorelPairing.lean`): To prove that the equal-rank pairing $R_F : \Omega_F^+ \to \Omega_F^-$ is jointly Borel measurable on $\mathcal{K} \times \mathbb{H}$ and induces a non-injective change-of-variables formula, the human paper cites the Lusin–Novikov measurable selection theorem or constructs contraction-mapping fixed points over rational discs. Instead, `BorelPairing.lean` combines Mathlib's inverse function theorem (`HasStrictFDerivAt.toOpenPartialHomeomorph`) with Second-Countable / Lindelöf exhaustion (`countable_joint_injective_partition`) and Polish-space measurable embeddings (`ContinuousOn.measurableEmbedding`) to construct a countable family of Borel partial inverses (`BorelBranches`) directly.

1.3. **Complete 47-Module Dependency Inventory.** Table 1 lists all 47 Lean 4 files in `OAI/Analysis/IntegralM` their line counts, and their mathematical roles.

Table 1: Complete inventory of the 47 Lean 4 modules in `OAI.Analysis.IntegralMeans`.

Section	Module	Lines	Primary Mathematical Content
§1 Specs	`Model.lean`	50	$\mathbb{D}, \mathcal{S}, M_t(f,r), \beta_f(t), B(t)$, `ConformalMapToDisk`
	`SharpModel.lean`	26	`koebeDomain`, `koebeInverse`, `areaMoment`
	`Main.lean`	20	Top-level `inverse_square_spectrum`, `brennan_conject`
	`SharpEndpoints.lean`	30	Top-level `sharp_schlicht_area`, `sharp_brennan_conje`
§2 Winding & Univalent Distortion	`LogIncrement.lean`	467	`HasLogIncrement`, `windingNumber`, `loopIndex`, `winding`
	`Turning.lean`	442	`secantTriangle`, `extendedSecant`, Hopf `tangent_turn`
	`Degree.lean`	704	`removeZero`, `surgeryLog`, `windingNumber_finite_regu`
	`UnivalentArea.lean`	262	Gronwall area theorem (`meromorphic_univalent_area`)
	`Distortion.lean`	984	$\lvert a_2\rvert \le 2$, Koebe 1/4, distortion, `schlicht_uniform_sepa`
§3 Compact Class & Transfer	`CompactClass.lean`	153	`DiskClass`, Arzelà–Ascoli compactness of $\mathcal{K}$
	`HalfPlane.lean`	419	$\mathbb{H}$, Cayley transforms, `rerootClass`, `classWeight` cocyc
	`TransferOperator.lean`	241	Dyadic operator L, subadditive spectral radius $\rho = 2^\beta$
	`CircleBounds.lean`	570	Cayley arcs, Proposition 2.4 (`uniform_inverse_square_`
	`Koebe.lean`	120	Koebe map $z/(1-z)^2 \in \mathcal{S}$, lower mean bound $\ge \frac{1}{4\pi(1-r)}$
§4 Eigenmeasure & Moments	`Eigenmeasure.lean`	476	Positive resolvent R_τ, weak-* eigenmeasure μ on $\mathcal{K}$
	`AffineLaw.lean`	609	Horizontal/scale averaging, Riesz law `HasWeightedLaw` I
	`Moments.lean`	505	Pre-Schwarzian $q_F = F''/F'$, $\mathbb{E}[a_F(i)]$, $\mathbb{E}[b_F(i)]$, $\mathbb{E}[\lvert q_F(i)\rvert$
§5 Critical Map & Escape	`CriticalMap.lean`	532	$G_F(z)$, $J_F(z)$, column rule, pointwise & expected J_F ide
	`InverseDensity.lean`	344	Inversion $z^\# = (-x+i)/y$, Lemma 4.2 ($\mathbb{E}[I(F)] \le 256\pi$)
	`BoundaryTail.lean`	339	Subharmonic L^2 ball mean, quartic boundary tail $o(y^3)$
	`TraceIdentity.lean`	531	`traceFlux` vector field, rectangular divergence identity
	`TraceBounds.lean`	272	Horizontal trace boundedness (`class_trace_bound`)
	`BoundaryEscape.lean`	454	Prop 4.1 (`compact_superlevels`), Lemma 4.3, `measural`
	`GoodPairs.lean`	311	Full measure of regular values, Lemma 5.4 (`ae_goodPai`
§6 Level Curves & Critical Count	`CriticalPoint.lean`	292	`regularizedGradient`, determinant at pole (> 0) & crit
	`Winding.lean`	461	`rectPath`, strip crossing jump $+1$ (`winding_jump_centr`
	`LevelCurve.lean`	515	Picard–Lindelöf flow of `tangentGradient`, closed level lo
	`LevelIndex.lean`	479	Maximum principle on `windingInterior`, `sum_loopInde`
	`CriticalCount.lean`	574	Lemma 5.5 (`GoodPair.card_maxima_eq_card_saddles:`

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Section	Module	Lines	Primary Mathematical Content
§7 Borel Pairing & Transport	`RankPairing.lean`	535	Lexicographic order `criticalPointLe`, `orderRank`, `crit`
	`BorelPairing.lean`	638	Lindelöf partition, `BorelBranches`, Prop 5.6 (`measurabl`
	`AreaTransport.lean`	236	`fiberSum`, non-injective change of variables (`transport_`
	`ExpectedDensity.lean`	118	`averagedDensity`, root test via scale-covariant law
	`TransportBoxes.lean`	220	`displacementWindow`, `transportBox`, `transport_box` ir
	`ExpectedTransport.lean`	396	Prop 6.1 ($\mathbb{E}[J_F(i)] \le 0$), Prop 6.2 ($\rho \le 2$), `transferBeta`
§8 Spectrum & Brennan's Conjecture	`Spectrum.lean`	105	Theorem 1.1: $\beta_f(-2) \le 1$, $\beta_{f_K}(-2) \ge 1$, $B(-2) = 1$
	`NegativeArea.lean`	271	Negative powers $t \in (-2, 0)$: `schlicht_negative_area`
	`SmoothMeans.lean`	312	Smoothed modulus $(1 + \|z\|^2)^q$, `schlicht_smooth_mean`
	`PositiveArea.lean`	168	Positive powers $t \in (0, 2/3)$: `schlicht_positive_area`
	`DomainArea.lean`	151	Full range $(-2, 2/3)$ & Brennan's Conjecture $(4/3, 4)$
§9 Sharp Endpoints	`AffineArea.lean`	60	Isometric affine invariance of `areaMoment`
	`KoebeReduction.lean`	72	Boundary derivative asymptotics of f_K near $z = \pm 1$
	`KoebeCone.lean`	52	Non-tangential sectorial cones at ± 1
	`KoebePolar.lean`	69	Polar coordinates in boundary cones, $1/r^2$ lower bounds
	`SectorDivergence.lean`	79	Divergence $\iint_{\text{sector}} c/r^2 \, dA = \infty$
	`KoebeDivergence.lean`	45	$M_{\text{area}}(f_K, -2) = \infty$ and $M_{\text{area}}(f_K, 2/3) = \infty$
	`InverseMoment.lean`	50	Endpoint divergence for φ_K at $p = 4/3$ and $p = 4$
Total	**47 Modules**	**14,759**	**Complete, 0-axiom, 0-sorry formalization**

2. Logarithmic Increments, Zero Surgery, and Classical Distortion (LogIncrement–Distortion)

2.1. **Logarithmic Increments and Winding Numbers (`LogIncrement.lean`).** Rather than defining winding numbers via contour integrals $\frac{1}{2\pi i} \oint \frac{\gamma'(t)}{\gamma(t)} \, dt$ (which requires piecewise C^1 regularity), `LogIncrement.lean` defines the winding number topologically for any continuous loop $\gamma \in C([0,1], \mathbb{C} \setminus \{0\})$ via covering-space lifts of $\exp : \mathbb{C} \to \mathbb{C}^\times$.

- **Logarithmic Increment and Winding Number** ($\gamma : C([0,1], \mathbb{C})$, $q, a \in \mathbb{C}$):

```
def HasLogIncrement (γ : C(unitInterval, ℂ)) (q : ℂ) : Prop :=
  ∃ T : C(unitInterval, ℂ), (∀ t, Complex.exp (T t) = γ t) ∧ T 1 - T 0 = q

def windingNumber (γ : C(unitInterval, ℂ)) : ℤ := by
  classical
  exact if h : γ 1 = γ 0 ∧ ∀ t, γ t ≠ 0 then
    Classical.choose (exists_integer_logIncrement γ h.1 h.2)
  else 0

def loopIndex (γ : C(unitInterval, ℂ)) (a : ℂ) : ℤ :=
  windingNumber ⟨fun t => γ t - a, by fun_prop⟩

def windingInterior (γ : C(unitInterval, ℂ)) : Set ℂ :=
  {a | a ∉ range γ ∧ loopIndex γ a ≠ 0}
```

Because $[0,1]$ is convex and hence contractible and locally path-connected, Mathlib's covering map lifting theorem (`Complex.isCoveringMapOn_exp.existsUnique_continuousMap_lifts`) guarantees a unique continuous lift $T : [0,1] \to \mathbb{C}$ with $e^{T(t)} = \gamma(t)$ (`exists_log_lift`). When $\gamma(1) = \gamma(0)$, $T(1) - T(0) = 2\pi i n$ for a unique integer $n = \text{wind}(\gamma) \in \mathbb{Z}$.

Any $\mathbb{R}$-linear map $A : \mathbb{C} \to_{\mathbb{R}} \mathbb{C}$ decomposes into complex-linear and conjugate-linear parts:

$$A(z) = \partial A \cdot z + \bar{\partial} A \cdot \bar{z}, \qquad \partial A = \frac{A(1) - iA(i)}{2}, \quad \bar{\partial} A = \frac{A(1) + iA(i)}{2}, \quad \det_{\mathbb{R}} A = |\partial A|^2 - |\bar{\partial} A|^2.$$

By Rouché perturbation (`HasLogIncrement.perturb`: if $\|\eta(t) - \gamma(t)\| < \|\gamma(t)\|$ for all t, then η/γ takes values in `Complex.slitPlane` and hence has a global logarithm $\log(\eta/\gamma)$), `local_logIncrement_pos` and `local_logIncrement_neg` prove that for a small circle $z_0 + re^{2\pi it}$ around an isolated zero $X(z_0) = 0$ with $DX(z_0) = A$, the local winding number is $+1$ if $\det_{\mathbb{R}} A > 0$ and -1 if $\det_{\mathbb{R}} A < 0$.

2.2. **Hopf Tangent Turning Theorem on Secant Triangles (`Turning.lean`).** For a C^1 simple closed curve $\gamma : \mathbb{R} \to \mathbb{C}$ of period 1 with $\gamma'(t) \neq 0$, `Turning.lean` proves that the tangent vector γ' has logarithmic increment $\pm 2\pi i$ without using piecewise polygonal approximations.

- **Secant Triangle and Extended Secant Map** ($\gamma : \mathbb{R} \to \mathbb{C}$, $p = (s, t) \in \mathbb{R}^2$):

```
def secantTriangle : Set (ℝ × ℝ) :=
  {p | 0 ≤ p.1 ∧ p.1 ≤ p.2 ∧ p.2 ≤ p.1 + 1}

def averagedDerivative (γ' : ℝ → ℂ) (p : ℝ × ℝ) : ℂ :=
  ∫ u in (0 : ℝ)..1, γ' (p.1 + u * (p.2 - p.1))

def extendedSecant (γ' : ℝ → ℂ) (p : ℝ × ℝ) : ℂ :=
  ((1 - (p.2 - p.1) : ℝ) : ℂ) * averagedDerivative γ' p -
    ((p.2 - p.1 : ℝ) : ℂ) * averagedDerivative γ' (p.1, p.2 - 1)
```

  By the fundamental theorem of calculus, for $0 < t - s < 1$,

$$\gamma(t) - \gamma(s) = (t - s)\,\texttt{averagedDerivative}(\gamma')(s, t),$$

  and by 1-periodicity $\gamma(t) = \gamma(t-1)$, so $\gamma(t) - \gamma(s)$ also equals $(1-(t-s))\,\texttt{averagedDerivative}(\gamma') - (t-s)\,\texttt{averagedDerivative}(\gamma')(s, t-1)$. Consequently, $\texttt{extendedSecant}(\gamma')(s, t)$ is **nowhere zero** on the entire convex triangle $\Delta = \{0 \le s \le t \le s + 1\}$:
  - on the diagonal $t = s$, $\texttt{extendedSecant}(s, s) = \gamma'(s) \neq 0$;
  - on the upper boundary $t = s + 1$, $\texttt{extendedSecant}(s, s + 1) = -\gamma'(s) \neq 0$;
  - on the interior $0 < t - s < 1$, $\texttt{extendedSecant}(s, t) = \frac{\gamma(t) - \gamma(s)}{t - s} \neq 0$ by injectivity of γ on $[0, 1)$.

  Since Δ is convex and hence contractible, `extendedSecant` admits a global continuous logarithm $L : \Delta \to \mathbb{C}$ (`exists_log_lift_of_simplyConnected`). Comparing the boundary paths from $(0, 0)$ to $(0, 1)$ via $(1, 1)$ versus directly along $(0, t)$ yields `tangent_turning_log`: if γ is positively oriented around an interior point a_0 ($\mathrm{Ind}(\gamma, a_0) = 1$), then $\mathrm{wind}(\gamma') = +1$ ($\Delta \log \gamma' = 2\pi i$).

2.3. **Poincaré–Hopf Degree Formula via Zero Surgery (`Degree.lean`).** To relate the winding number of a vector field $X : \mathbb{C} \to \mathbb{C}$ along a closed loop γ inside a convex set $K \subset \mathbb{C}$ to its finite set $s \subset K$ of regular zeros, `Degree.lean` performs **multiplicative zero surgery**:

- **Zero Surgery Construction and Degree Sum Theorem** ($X : C(\mathbb{C}, \mathbb{C})$, $s : \mathrm{Finset}\ \mathbb{C}$, $K \subseteq \mathbb{C}$):

```
def planeIndex (A : ℂ →L[ℝ] ℂ) : ℤ :=
  if 0 < LinearMap.det A.toLinearMap then 1 else -1

def zeroModel (a : ℂ) (A : ℂ →L[ℝ] ℂ) (z : ℂ) : ℂ :=
  if 0 < LinearMap.det A.toLinearMap then
    linearPart A * (z - a)
  else
    antiLinearPart A * conj (z - a)

lemma windingNumber_finite_regular_zeros {K : Set ℂ} (hK : Convex ℝ K)
```

```
(s : Finset ℂ) (X : C(ℂ, ℂ)) (A : ℂ → (ℂ →L[ℝ] ℂ))
(hsK : ∀ a ∈ s, a ∈ K)
(hzero : ∀ z ∈ K, X z = 0 ↔ z ∈ s)
(hreg : ∀ a ∈ s, HasFDerivAt X (A a) a ∧ LinearMap.det (A a).toLinearMap ≠ 0)
(γ : C(unitInterval, ℂ)) (hloop : γ 1 = γ 0) (hγK : ∀ t, γ t ∈ K)
(hγ0 : ∀ t, X (γ t) ≠ 0) :
windingNumber (X.comp γ) = ∑ a ∈ s, planeIndex (A a) * loopIndex γ a
```

Proof Mechanism (`removeZero` and `surgeryLog`): Near a regular zero $a \in s$, the ratio $X(z)/\texttt{zeroModel}(a, A_a, z)$ tends to $1 + \frac{\bar\partial A_a}{\partial A_a}\frac{\bar z - \bar a}{z-a}$ (if $\det_{\mathbb{R}} A_a > 0$), which lies strictly in the right half-plane $\operatorname{Re} w > 0 \subset \texttt{slitPlane}$ for $0 < \|z-a\| < 2r$ because $|\bar\partial A_a| < |\partial A_a|$. Using a smooth radial cutoff χ supported in $B(a,2r)$ with $\chi \equiv 1$ on $B(a,r)$, `removeZero` defines

$$\widetilde{X}(z) := \begin{cases} \exp\Big((1-\chi(z))\log\frac{X(z)}{\texttt{zeroModel}(a,A_a,z)}\Big) & \text{if } \|z-a\| < 2r, \\ \dfrac{X(z)}{\texttt{zeroModel}(a,A_a,z)} & \text{if } \|z-a\| \geq r. \end{cases}$$

Then $\widetilde{X}$ is continuous on all of $\mathbb{C}$, has zero set $s \setminus \{a\}$ on K, and along γ (after shrinking r so γ avoids $B(a,2r)$) satisfies $X(\gamma(t)) = \texttt{zeroModel}(a, A_a, \gamma(t))\,\widetilde{X}(\gamma(t))$, so

$$\operatorname{wind}(X \circ \gamma) = \texttt{planeIndex}(A_a)\,\operatorname{Ind}(\gamma, a) + \operatorname{wind}(\widetilde{X} \circ \gamma).$$

Inducting on $\#s$ reduces to $s = \emptyset$, where X is nonvanishing on convex K and hence $\operatorname{wind}(X \circ \gamma) = 0$.

As an immediate application, `Degree.lean` proves `integral_jacobian_nonpos_of_boundary_winding`: if a C^1 map $X : \mathbb{C} \to \mathbb{C}$ has boundary winding number $\operatorname{wind}(X(re^{2\pi it}) - \xi) \leq 0$ for every $\xi \notin X(\partial B(0,r))$, then at every regular value ξ with a preimage in $B(0,r)$, the sum of indices of preimages in $B(0,r)$ is ≤ 0, so every preimage with $\det_{\mathbb{R}} DX(z) > 0$ must be accompanied by at least one preimage with $\det_{\mathbb{R}} DX(w) < 0$. Applying area change of variables on positive and negative Jacobian regions yields $\int_{\overline{B}(0,r)} \det_{\mathbb{R}} DX(z)\,dA(z) \leq 0$.

2.4. **Gronwall Area Theorem and Univalent Distortion (`UnivalentArea.lean`, `Distortion.lean`).** In `UnivalentArea.lean`, the topological degree inequality $\int_{\overline{B}(0,r)} \det_{\mathbb{R}} DX \leq 0$ is applied to the smooth extension $X(z) = r^{-2}\bar z + h(z)$ of a univalent meromorphic function $H(z) = z^{-1} + h(z)$ on $\mathbb{D} \setminus \{0\}$ (which coincides with $H(z)$ on the circle $\|z\| = r$ since $z^{-1} = r^{-2}\bar z$ there). Since $\det_{\mathbb{R}} DX(z) = |h'(z)|^2 - r^{-4}$, this gives Gronwall's area inequality (`meromorphic_univalent_area`):

$$\int_{\overline{B}(0,r)} |h'(z)|^2\,dA(z) \leq \pi r^2 \cdot r^{-4} = \pi r^{-2}.$$

In `Distortion.lean`, combining Gronwall's area inequality with the mean-value inequality $\pi r^2 |h'(0)|^2 \leq \int_{\overline{B}(0,r)} |h'(z)|^2\,dA$ and letting $r \to 1^-$ gives $|h'(0)| \leq 1$ (`meromorphic_univalent_coefficient`). Applying the square-root transform $H(z) = 1/\sqrt{f(z^2)}$ then yields:

(1) **Bieberbach's Second Coefficient Bound** (`schlicht_second_deriv_bound`): $|f''(0)| \leq 4$ for all $f \in \mathcal{S}$.
(2) **Koebe** $1/4$ **Theorem** (`koebe_quarter`): $B(0, 1/4) \subseteq f(\mathbb{D})$ for all $f \in \mathcal{S}$.
(3) **Pre-Schwarzian and Distortion Bounds** (`schlicht_preSchwarzian_bound`, `schlicht_deriv_bounds`, `schlicht_growth_bounds`): for all $f \in \mathcal{S}$ and $z \in \mathbb{D}$,

$$\left|\frac{f''(z)}{f'(z)} - \frac{2\bar z}{1-\|z\|^2}\right| \leq \frac{4}{1-\|z\|^2}, \qquad \frac{1-\|z\|}{(1+\|z\|)^3} \leq |f'(z)| \leq \frac{1+\|z\|}{(1-\|z\|)^3}, \qquad \frac{\|z\|}{(1+\|z\|)^2} \leq |f(z)| \leq \frac{\|z\|}{(1-\|z\|)^2}.$$

(4) **Uniform Separation Lemma** (`schlicht_uniform_separation`): for any $z, w \in \mathbb{D}$ with $z \neq w$, letting $d = 1 - \|z\|$ and $\delta = \min(d/2, \|w - z\|/2)$, the disk re-normalization $g(u) = \frac{f(z+\delta u)-f(z)}{\delta f'(z)} \in \mathcal{S}$ omits $\frac{f(w)-f(z)}{\delta f'(z)}$ since $w \notin B(z, \delta)$, so by Koebe 1/4,

$$\|f(w) - f(z)\| \geq \frac{\delta}{4}|f'(z)| \geq \frac{\min(1 - \|z\|, \|w - z\|)\,(1 - \|z\|)}{8(1 + \|z\|)^3} > 0,$$

giving a strictly positive lower bound independent of $f \in \mathcal{S}$.

3. The Compact Class $\mathcal{K}$ and the Dyadic Transfer Operator (CompactClass–CircleBounds)

3.1. Compactness of `DiskClass` via Arzelà–Ascoli (`CompactClass.lean`).

- **Compact Schlicht Class** (`diskSchlichtSet` $\subseteq C(\mathbb{D}, \mathbb{C})$):

```
def diskExtend (g : C(disk, ℂ)) (z : ℂ) : ℂ :=
  if hz : z ∈ disk then g ⟨z, hz⟩ else 0

def diskSchlichtSet : Set C(disk, ℂ) :=
  {g : C(disk, ℂ) | Schlicht (diskExtend g)}

abbrev DiskClass : Type := diskSchlichtSet
```

Theorem 3.1 (`isCompact_diskSchlichtSet`, `CompactSpace DiskClass`)**.** *Equipped with the topology of uniform convergence on compact subsets of* $\mathbb{D}$*,* ***`diskSchlichtSet`*** *is closed and compact in* $C(\mathbb{D}, \mathbb{C})$*.*

Proof. For each compact $K \subset \mathbb{D}$, choosing $R \in (0, 1)$ with $K \subseteq \overline{B}(0, R)$, the distortion bounds give $\|g(z)\| \leq R/(1 - R)^2$ on K and Lipschitz constant $(1 + R)/(1 - R)^3$ on $\overline{B}(0, R)$. By `ArzelaAscoli.isCompact_closure_of_isClosedEmbedding`, the closure of `diskSchlichtSet` in $C(\mathbb{D}, \mathbb{C})$ is compact. To prove `diskSchlichtSet` is closed (`isClosed_diskSchlichtSe`

(i) Cauchy's integral formula on closed balls (`Complex.cauchyIntegral_circleMap`) passes through uniform limits on compacta to preserve complex differentiability and derivatives at 0.

(ii) Injectivity is preserved in the limit by `schlicht_uniform_separation`: if $g_n \to g$ pointwise and $z \neq w \in \mathbb{D}$, then $\|g_n(w) - g_n(z)\| \geq c(z, w) > 0$ for all n, so $\|g(w) - g(z)\| \geq c(z, w) > 0$. □

3.2. Upper Half-Plane Models and the Affine Cocycle (`HalfPlane.lean`).

The Cayley transforms $\tau_{\mathbb{D}}(z) = \frac{z-i}{z+i} : \mathbb{H} \to \mathbb{D}$ (`diskCayley`) and $\tau_{\mathbb{H}}(w) = i\frac{1+w}{1-w} : \mathbb{D} \to \mathbb{H}$ (`halfCayley`) induce a bijection between $\mathcal{S}$ and the upper half-plane schlicht class $\mathcal{K} = \{F : \mathbb{H} \to \mathbb{C} \text{ univalent} \mid F(i) = 0,\ F'(i) = 1\}$ via $F(z) = 2i\, f(\tau_{\mathbb{D}}(z))$ (`diskToHalf`).

- **Half-Plane Embedding, Affine Rerooting, and Cocycle Weight** ($g \in$ `DiskClass`, $z, w \in \mathbb{H}$):

```
def halfPlane : Set ℂ := {z | 0 < z.im}

def affine (z w : ℂ) : ℂ := (z.re : ℂ) + (z.im : ℂ) * w

def reroot (f : ℂ → ℂ) (z w : ℂ) : ℂ :=
  (f (affine z w) - f z) / ((z.im : ℂ) * deriv f z)

def reciprocalDeriv (f : ℂ → ℂ) (z : ℂ) : ℂ := (deriv f z)⁻¹

def classFun (g : DiskClass) : ℂ → ℂ := diskToHalf (diskExtend g.1)
```

```
def rerootClass (g : DiskClass) (z : halfPlane) : DiskClass :=
  classOfHalf (reroot (classFun g) z) (reroot_normalized (classFun_schlicht g).1 z.2)

def classWeight (g : DiskClass) (z : halfPlane) : ℝ :=
  ‖reciprocalDeriv (classFun g) z‖ ^ 2
```

In mathematical notation, for $F = \texttt{classFun}(g) \in \mathcal{K}$ and $z = x + iy \in \mathbb{H}$, the affine map $\alpha_z(w) = x + yw$ defines a group structure on $\mathbb{H}$ ($\alpha_z \circ \alpha_w = \alpha_{\alpha_z(w)}$, `halfMul`). The re-rooted map $F_z(w) = \frac{F(x+yw)-F(z)}{yF'(z)} \in \mathcal{K}$ (`rerootClass`) and the reciprocal derivative $H_F(z) = 1/F'(z)$ (`reciprocalDeriv`) satisfy the exact group action and multiplicative cocycle identities (`rerootClass_mul`, `classWeight_cocycle`):

$$(F_z)_w = F_{\alpha_z(w)}, \qquad H_{F_z}(w) = \frac{H_F(\alpha_z(w))}{H_F(z)}, \qquad \omega(g, \alpha_z(w)) = \omega(g, z)\,\omega(\texttt{rerootClass}(g, z), w),$$

where $\omega(g, z) := |H_F(z)|^2 = |F'(z)|^{-2}$ (`classWeight`). Moreover, the disk distortion bound translates via Cayley transform to the half-plane reciprocal bound (`classFun_reciprocal_bound`):

$$|H_F(z)| \le \frac{2\|z + i\|^4}{y}, \qquad |H_F'(z)| \le \frac{6}{y}|H_F(z)|.$$

3.3. The Dyadic Transfer Operator and Disk Means Comparison (`TransferOperator.lean`, `CircleBounds.lean`).

- **Affine Action Operator, Dyadic Transfer Operator L, and Growth Rate ρ** ($z \in \mathbb{H}$, $\Phi \in C(\texttt{DiskClass}, \mathbb{R})$):

```
def affineOperator (z : halfPlane) : C(DiskClass, ℝ) →L[ℝ] C(DiskClass, ℝ)

def transferStep (k : Fin 2) : halfPlane := halfPoint k (1 / 2) (by norm_num)

def transferOperator : C(DiskClass, ℝ) →L[ℝ] C(DiskClass, ℝ) :=
  affineOperator (transferStep 0) + affineOperator (transferStep 1)

def growthLog : ℝ :=
  (transferOperator_subadditive_log.lim : ℝ)

def transferRho : ℝ := Real.exp growthLog

def transferBeta : ℝ := growthLog / Real.log 2
```

Mathematically, for each $z \in \mathbb{H}$, $(T_z\Phi)(F) = |H_F(z)|^2\Phi(F_z)$ is a positive continuous linear operator on $C(\mathcal{K}, \mathbb{R})$ satisfying $T_{\alpha_z(w)} = T_z \circ T_w$. The dyadic transfer operator is

$$(L\Phi)(F) := (T_{i/2}\Phi)(F) + (T_{1+i/2}\Phi)(F) = |H_F(i/2)|^2\Phi(F_{i/2}) + |H_F(1 + i/2)|^2\Phi(F_{1+i/2}).$$

By induction (`transferOperator_pow_apply`), for every $n \ge 1$,

$$(L^n\Phi)(F) = \sum_{j=0}^{2^n-1} |H_F(j2^{-n} + i2^{-n})|^2\, \Phi(F_{j2^{-n}+i2^{-n}}).$$

Since $a_n := \log \|L^n\|$ is subadditive ($a_{m+n} \le a_m + a_n$, `transferOperator_subadditive_log`) and bounded below by $n \log c_0$ (using $|H_F(i/2)|^2 > 0$ on compact $\mathcal{K}$), Fekete's subadditive lemma (`Subadditive.tendsto_lim`) gives $\lim_{n\to\infty} \frac{1}{n} \log \|L^n\| = \log \rho$ with $\rho = 2^\beta > 0$ (`transferRho_eq_two_rpow`).

- **Proposition 2.4: Uniform Disk Comparison (`CircleBounds.lean`)**: Under $w = -re^{i\theta}$ for $r \in [1/2, 1)$ and $\theta \in [-\pi/4, \pi/4]$, the Cayley image $\tau_{\mathbb{H}}(-re^{i\theta}) = X(r, \theta) + iY(r, \theta)$ satisfies $X(r, \theta) \in [-1, 1]$, $\frac{1-r}{2} \le Y(r, \theta) \le 2(1 - r)$, and $\partial_\theta X(r, \theta) \ge 1/16$ (`arcXDeriv_lower`).

Comparing horizontal integrals at height $Y(r,\theta)$ to dyadic sums at height $2^{-n} \asymp 1-r$ via `classWeight_vertical_compare` and covering $[-\pi,\pi]$ by four rotated arcs gives `uniform_inverse_square_bound`: for every $\varepsilon > 0$, there exists $C_\varepsilon > 0$ such that for all $f \in \mathcal{S}$ and $r \in [1/2, 1)$,

$$M_{-2}(f,r) \le C_\varepsilon (1-r)^{-\beta-\varepsilon} \quad \text{whenever } \rho \le 2.$$

4. Eigenmeasure, Scale-Covariant Law, and Pre-Schwarzian Moments (Eigenmeasure–Moments)

4.1. Eigenmeasure via Positive Resolvent and Weak-* Compactness (`Eigenmeasure.lean`).

- **Positive Resolvent and Weak-* Eigenmeasure** ($\tau > \rho$, $f \in C(\texttt{DiskClass}, \mathbb{R})$):

```
def positiveResolvent (τ : ℝ) : C(DiskClass, ℝ) →L[ℝ] C(DiskClass, ℝ) :=
  ∑' n : ℕ, (τ⁻¹ ^ n) • (transferOperator ^ n)

lemma exists_transfer_eigenmeasure :
    ∃ μ : ProbabilityMeasure DiskClass,
      ∀ f : C(DiskClass, ℝ),
        ∫ g, transferOperator f g ∂(μ : Measure DiskClass) =
          transferRho * ∫ g, f g ∂(μ : Measure DiskClass)
```

Proof Mechanism: For each $\tau > \rho$, the series $R_\tau = \sum_{n=0}^{\infty} \tau^{-n} L^n$ converges in operator norm, satisfies $L(R_\tau f) = \tau(R_\tau f - f)$ (`transferOperator_positiveResolvent`), and maps 1 to a strictly positive function $R_\tau 1 \ge 1$. Since $\mathcal{K}$ is compact, $R_\tau 1$ attains its maximum at some $g_\tau \in \mathcal{K}$ (`exists_maximizer`), and because $\limsup_{n\to\infty} \|L^n 1\|^{1/n} = \rho$, we have $\lim_{\tau\downarrow\rho}(R_\tau 1)(g_\tau) = +\infty$ (`tendsto_max_positiveResolvent_one_atTop`). Defining the normalized positive linear functional

$$\Lambda_\tau(f) := \frac{(R_\tau f)(g_\tau)}{(R_\tau 1)(g_\tau)},$$

Riesz representation (`ProbabilityMeasure`) realizes Λ_τ as a probability measure $\mu_\tau \in \mathcal{P}(\mathcal{K})$ satisfying

$$\left| \int_{\mathcal{K}} (Lf)\, d\mu_\tau - \tau \int_{\mathcal{K}} f\, d\mu_\tau \right| = \frac{\tau |f(g_\tau)|}{(R_\tau 1)(g_\tau)} \le \frac{\tau \|f\|_\infty}{(R_\tau 1)(g_\tau)} \xrightarrow{\tau\downarrow\rho} 0.$$

Since $\mathcal{P}(\mathcal{K})$ is a compact space in Mathlib (`CompactSpace (ProbabilityMeasure DiskClass)`), the net μ_τ has a weak-* cluster point $\mu \in \mathcal{P}(\mathcal{K})$ as $\tau \downarrow \rho$, which satisfies $L^*\mu = \rho\mu$.

4.2. Horizontal and Logarithmic Scale Averaging (`AffineLaw.lean`).

Assuming $\rho > 2$ (i.e., $\beta = \log_2 \rho > 1$), `AffineLaw.lean` upgrades the discrete dyadic eigenmeasure μ ($L^*\mu = 2^\beta \mu$) to a probability measure $P \in \mathcal{P}(\mathcal{K})$ covariant under the *entire continuous* affine group $\mathbb{H}$:

- **Continuous Scale-Covariant Law** ($P \in \mathcal{P}(\texttt{DiskClass})$, $\beta \in \mathbb{R}$):

```
def HasWeightedLaw (P : ProbabilityMeasure DiskClass) (β : ℝ) : Prop :=
  ∀ z : halfPlane, ∀ f : C(DiskClass, ℝ),
    ∫ g, affineOperator z f g ∂(P : Measure DiskClass) =
      (z.val.im) ^ (-β) * ∫ g, f g ∂(P : Measure DiskClass)

lemma exists_weighted_continuous_law (hρ : 2 < transferRho) :
    ∃ P : ProbabilityMeasure DiskClass, HasWeightedLaw P transferBeta
```

Two-Step Averaging Construction:

(1) **Horizontal Averaging**: Define $\bar{\mu}(f) := \frac{1}{2}\int_0^2 \left(\int_{\mathcal{K}} (T_{u+i}f)\, d\mu\right) du$ (`horizontalAverage`). Using $L^*\mu = 2^\beta \mu$ and $T_{j+i/2} = T_{i/2+j/2}$ on $[0,2]$, one checks the exact dyadic scaling law $\bar{\mu}(T_{i/2}f) = 2^{\beta-1}\bar{\mu}(f)$ (`horizontalAverage_halfScale`) and shows that for $t > 0$, $\bar{\mu}(T_{x+it}f) - \bar{\mu}(T_{it}f)$ is $2t$-periodic in x and vanishes as $t \to \infty$ (since $\beta > 1$ implies $2^{n(1-\beta)} \to 0$).

(2) **Logarithmic Scale Averaging**: Averaging $t^{\beta-1}\bar{\mu}(T_{it}f)$ over $t \in [1,2]$ with respect to $\frac{dt}{t \log 2}$ yields a probability measure $P \in \mathcal{P}(\mathcal{K})$ that is both horizontally translation-invariant and scale-covariant of exponent $-\beta$: for all $z = x + iy \in \mathbb{H}$ and all Borel measurable $f : \mathcal{K} \to [0,\infty]$ (`exists_weighted_law` in `ExpectedDensity.lean`),

$$\mathbb{E}_P\big[|H_F(z)|^2 f(F_z)\big] = y^{-\beta}\mathbb{E}_P[f(F)].$$

4.3. Pre-Schwarzian Moments at $z = i$ (`Moments.lean`).

- **Pre-Schwarzian Coefficients and Moment Identities** ($f : \mathbb{C} \to \mathbb{C}$, $z \in \mathbb{C}$):

```
def preSchwarzian (f : ℂ → ℂ) (z : ℂ) : ℂ := deriv (deriv f) z / deriv f z
def coeffA (f : ℂ → ℂ) (z : ℂ) : ℝ := (z.im : ℝ) * (preSchwarzian f z).im
def coeffB (f : ℂ → ℂ) (z : ℂ) : ℝ := (z.im : ℝ) * (preSchwarzian f z).re
```

Writing $q_F(z) = F''(z)/F'(z)$, $a_F(z) = y \operatorname{Im} q_F(z)$, and $b_F(z) = y \operatorname{Re} q_F(z)$, the re-rooting transformation gives $q_{F_z}(i) = y q_F(z)$, so $a_{F_z}(i) = a_F(z)$, $b_{F_z}(i) = b_F(z)$, and $|q_{F_z}(i)|^2 = y^2|q_F(z)|^2$. Differentiating the scale-covariance identity $\Psi(z) := \mathbb{E}_P[|H_F(z)|^2] = y^{-\beta}$ at $z = i$ using $\partial_x|H_F|^2 = -2|H_F|^2 \operatorname{Re} q_F$, $\partial_y|H_F|^2 = 2|H_F|^2 \operatorname{Im} q_F$, and $\Delta|H_F|^2 = 4|H_F'(z)|^2 = 4|H_F|^2|q_F|^2$ produces the three exact moment identities (`expected_coeffA`, `expected_coeffB`, `expected_preSchwarzian_norm_sq`):

$$\mathbb{E}_P[a_F(i)] = -\frac{\beta}{2}, \qquad \mathbb{E}_P[b_F(i)] = 0, \qquad \mathbb{E}_P\big[|q_F(i)|^2\big] = \frac{\beta(\beta+1)}{4}.$$

5. The Critical Map, Quadratic Positivity, and Boundary Escape (CriticalMap–GoodPairs)

5.1. The Critical Map G_F and the Normalized Jacobian Identity (`CriticalMap.lean`).

Fix a parameter $k > 0$ (later specialized to $k = (\beta+3)/4$). For $F \in \mathcal{K}$ and $\xi \in \mathbb{C}$, consider the reciprocal potential $V_\xi^F(z) = y^k|F(z) - \xi|^{-1}$. Its logarithmic critical points in $\mathbb{H} \setminus \{F^{-1}(\xi)\}$ are solved explicitly by $\xi = G_F(z)$, where:

- **Critical Map and Normalized Jacobian** ($f : \mathbb{C} \to \mathbb{C}$, $k \in \mathbb{R}$, $z \in \mathbb{H}$):

```
def criticalMap (f : ℂ → ℂ) (k : ℝ) (z : ℂ) : ℂ :=
  f z + (Complex.I / (k : ℂ)) * (z.im : ℂ) * deriv f z

def normalizedJacobian (f : ℂ → ℂ) (k : ℝ) (z : ℂ) : ℝ :=
  LinearMap.det (fderiv ℝ (criticalMap f k) z).toLinearMap / ‖deriv f z‖ ^ 2
```

Mathematically,

$$G_F(z) := F(z) + \frac{iy}{k}F'(z), \qquad J_F(z) := \frac{\det_{\mathbb{R}} DG_F(z)}{|F'(z)|^2}.$$

Computing the real partial derivatives of G_F at $z = x+iy \in \mathbb{H}$ (`criticalMap_hasFDerivAt`):

$$\partial_x G_F(z) = F'(z)\left(1 + \frac{iy}{k}q_F(z)\right), \qquad \partial_y G_F(z) = iF'(z)\left(\frac{k-1}{k} + \frac{iy}{k}q_F(z)\right).$$

Factoring out $F'(z)$ (which scales determinants by $|F'(z)|^2$) yields the fundamental algebraic identity (`normalizedJacobian_identity`):

$$k^2 J_F(z) = k(k-1) + (2k-1)a_F(z) + |y q_F(z)|^2.$$

Taking expectations at $z = i$ using $\mathbb{E}_P[a_F(i)] = -\beta/2$ and $\mathbb{E}_P[|q_F(i)|^2] = \beta(\beta+1)/4$ gives (`expected_normalizedJacobian`):

$$k^2\mathbb{E}_P[J_F(i)] = k(k-1) - \frac{\beta(2k-1)}{2} + \frac{\beta(\beta+1)}{4}.$$

Specializing to $k = (\beta+3)/4$ (`criticalK`), the right-hand side simplifies algebraically (`critical_polynomial_identity`) to

$$k^2\mathbb{E}_P[J_F(i)] = \frac{(\beta-1)(3-\beta)}{16}.$$

Therefore (`expected_normalizedJacobian_pos`): **for every** $\beta \in (1,3)$**,** $\mathbb{E}_P[J_F(i)] > 0$**!**

5.2. **Weighted Inverse Area and Quartic Boundary Tail (`InverseDensity.lean`, `BoundaryTail.lean`).** To show that for P-a.e. $F \in \mathcal{K}$ and $k = (\beta+3)/4$, the potential $V_\xi^F(z) = y^k/|F(z) - \xi|$ vanishes as $z \to \partial\mathbb{H} \cup \{\infty\}$, `InverseDensity.lean` and `BoundaryTail.lean` establish:

(1) **Weighted Inverse Area Bound (Lemma 4.2, `weightedInverseArea_lintegral_le`)**: Define the core $B_0 = \overline{B}(i, 1/2)$ and the exterior $S_0 = \mathbb{H} \setminus B_0$ (`outsideCore`). On S_0, Koebe's 1/4 theorem transferred to $\mathbb{H}$ gives $|F(z)| \geq 1/16$ (`classFun_norm_ge_outsideCore`), so the injective map $w = 1/F(z)$ maps S_0 into $\overline{B}(0,16)$ with area $\int_{S_0} |F'(z)|^2 |F(z)|^{-4}\, dA(z) \leq 256\pi$ (`inverse_area_bound`). Applying the involutive inversion $z^\# = \frac{-x+i}{y}$ (`affineInverse`, with real Jacobian determinant y^{-3} and $F_z(z^\#) = \frac{-F(z)}{yF'(z)}$) and integrating against P using scale-covariance $\mathbb{E}_P[|H_F(z)|^2 I(F_z)] = y^{-\beta}\mathbb{E}_P[I(F)]$ gives

$$\mathbb{E}_P\left[\int_{S_0} y^{\beta-1}|F(z)|^{-4}\, dA(z)\right] \leq 256\pi < \infty.$$

(2) **Quartic Boundary Tail (`BoundaryTail.lean`, `HasQuarticTail`)**: For any F with $I_\alpha(F) := \int_{S_0} y^{\alpha-4}|F(z)|^{-4}\, dA(z) < \infty$ (where $\alpha = \beta+3 = 4k$), applying the subharmonic L^2 mean-value inequality (`holomorphic_sq_submean_lintegral`) to the holomorphic function $q(w) = F(w)^{-2}$ on the disk $B(z, y/4) \subset S_0 \setminus K$ yields

$$y^\alpha |F(z)|^{-4} \leq \frac{4^{\alpha-2}}{\pi} \int_{B(z,y/4)} \eta^{\alpha-4}|F(w)|^{-4}\, dA(w) \xrightarrow{z \to \partial\mathbb{H}\cup\{\infty\}} 0.$$

Taking fourth roots gives $\sup_{z\in\mathbb{H}\setminus K} y^k|F(z)|^{-1} \to 0$, so every positive superlevel $\{z \in \mathbb{H} : V_\xi^F(z) \geq c\}$ $(c > 0)$ is compact in $\mathbb{H}$ (`compact_superlevels`, Proposition 4.1).

5.3. **The Rectangular Divergence Trace Identity and Boundary Escape (`TraceIdentity.lean`–`GoodPairs.lean`).** To extend $I_\alpha(F) < \infty$ from $\xi = 0$ to Lebesgue-almost every $\xi \in \mathbb{C}$, we need the horizontal trace bound $\sup_{0<y\leq 1} \int_{\mathbb{R}} (1 + |F(x+iy) - \xi|^2)^{-2}\, dx < \infty$ for a.e. $\xi \in \mathbb{C}$.

- **Rectangular Divergence Trace Identity (`TraceIdentity.lean`)**:

```
def traceFluxX (ε : ℝ) (g : ℝ × ℝ → ℝ) (p : ℝ × ℝ) : ℝ :=
  (p.2 - ε) * fderiv ℝ g p (1, 0)

def traceFluxY (ε : ℝ) (g : ℝ × ℝ → ℝ) (p : ℝ × ℝ) : ℝ :=
  (p.2 - ε) * fderiv ℝ g p (0, 1) - g p

lemma divergence_traceFlux (ε : ℝ) {g : ℝ × ℝ → ℝ} {p : ℝ × ℝ}
    (hg : ContDiffAt ℝ 2 g p) :
    fderiv ℝ (traceFluxX ε g) p (1, 0) + fderiv ℝ (traceFluxY ε g) p (0, 1) =
      (p.2 - ε) * (secondReal (g ∘ Complex.equivRealProdCLM) (p.1 + p.2 * Complex.I) 1 1 +
        secondReal (g ∘ Complex.equivRealProdCLM) (p.1 + p.2 * Complex.I) Complex.I Complex.
    I)
```

Bypass of Differentiating Under the Integral Sign: Let $\chi \in C_c^\infty(\mathbb{C})$ be a smooth bump (`exists_trace_smooth_cutoff`) with $\chi \equiv 1$ on $[-1,1]\times[0,1]$ and $\operatorname{supp}\chi \subseteq [-2,2]\times[-2,2]$, let $\psi \in C^2(\mathbb{C},\mathbb{R})$ be nonnegative, and set $g(x,y) = \chi(x,y)\,\psi(F(x+iy))$. On the rectangle $[-2,2]\times[\varepsilon,2]$ $(0<\varepsilon\le 1)$, g vanishes on the left, right, and top boundaries, while on the bottom boundary $y=\varepsilon$, `traceFluxY`$(\varepsilon,g)(x,\varepsilon) = -g(x,\varepsilon)$. Therefore, Mathlib's divergence theorem on rectangles (`integral_divergence_prod_Icc_of_hasFDerivAt_of_le`) immediately yields `rectangle_trace_identity`:

$$\int_{-1}^{1} \psi(F(x+i\varepsilon))\,dx \le \int_{-2}^{2} g(x,\varepsilon)\,dx = \int_{\varepsilon}^{2}\int_{-2}^{2} (y-\varepsilon)\Delta g(x,y)\,dx\,dy.$$

Expanding $\Delta(\chi\cdot(\psi\circ F)) = (\Delta\chi)(\psi\circ F) + 2\nabla\chi\cdot\nabla(\psi\circ F) + \chi|F'|^2(\Delta\psi)\circ F$, bounding $y \le C|F'(z)|$ via Koebe distortion, and changing variables $w = F(z)$ via `conformal_lintegral_le` bounds the horizontal trace uniformly for $\varepsilon \in (0,1]$ (`class_trace_bound` in `TraceBounds.lean`).

- **Boundary Escape and Full Measure of Good Pairs (`BoundaryEscape.lean`, `GoodPairs.lean`)**: Integrating $(1+|F(z)-\xi|^2)^{-2}$ over $\xi \in B(0,R)$ by Tonelli's theorem and combining with the exterior bound gives $\int_{\mathbb{H}} y^{\beta-1}(1+|F(z)-\xi|^2)^{-2}\,dA(z) < \infty$ for Lebesgue-a.e. $\xi\in\mathbb{C}$, which implies `HasQuarticTail` for $F-\xi$ (`ae_shifted_quarticTail`, Lemma 4.3). Meanwhile, `ae_regular_values` in `GoodPairs.lean` applies Mathlib's `addHaar_image_eq_zero_of_det_fderivWithi` on $\mathbb{H}$ to show that the critical values $G_F(\{z\in\mathbb{H} : J_F(z)=0\})$ have Lebesgue measure zero. Thus, for P-a.e. $F\in\mathcal{K}$, almost every $\xi\in\mathbb{C}$ is a **good pair** (`GoodPair`): $F-\xi$ has compact positive superlevels of V_ξ^F and ξ is a regular value of G_F (`ae_goodPairs_of_escape`, Lemma 5.4).

6. Regularized Gradients, Level Loops, and the Critical Point Count (CriticalPoint–CriticalCount)

6.1. **The Regularized Gradient and Elimination of the Pole (`CriticalPoint.lean`).** Fix a good pair (F,ξ) and $k>0$. Instead of $\ell(z) = \log V_\xi^F(z)$ (which blows up at $F(z)=\xi$), `CriticalCount.lean` studies the globally C^∞ **squared reciprocal potential**

$$u(z) := \frac{|F(z)-\xi|^2}{y^{2k}} = \big(V_\xi^F(z)\big)^{-2} \qquad (\texttt{squaredReciprocalPotential}),$$

so that a superlevel $\{V_\xi^F \ge \Lambda\}$ is a sublevel $\{u\le c\}$ with $c=\Lambda^{-2}>0$. On $\{u>0\}$, a direct computation (`squaredReciprocalPotential_hasFDerivAt`) shows that the Euclidean gradient of u is

$$\nabla u(z) = -\frac{2k}{y^{2k}}\cdot\frac{ik}{y}(F(z)-\xi)\overline{(G_F(z)-\xi)}.$$

Stripping off the positive factor $2ky^{-2k}$ defines the **regularized gradient**:

- **Regularized Gradient and Determinant Identities** ($f:\mathbb{C}\to\mathbb{C}$, $k\in\mathbb{R}$, $\xi,z\in\mathbb{C}$):

```
def regularizedGradient (f : ℂ → ℂ) (k : ℝ) (ξ z : ℂ) : ℂ :=
  (Complex.I * (k : ℂ) / (z.im : ℂ)) * (f z - ξ) * conj (criticalMap f k z - ξ)

lemma regularizedGradient_eq_zero {f : ℂ → ℂ} {k : ℝ} (hk : k ≠ 0) {ξ z : ℂ}
    (hz : z ∈ halfPlane) :
    regularizedGradient f k ξ z = 0 ↔ f z = ξ ∨ criticalMap f k z = ξ

lemma det_regularizedGradient_at_pole_pos {f : ℂ → ℂ} (hf : UnivalentOn f halfPlane)
    {k : ℝ} (hk : k ≠ 0) {ξ z : ℂ} (hz : z ∈ halfPlane) (hpole : f z = ξ) :
    0 < LinearMap.det (fderiv ℝ (regularizedGradient f k ξ) z).toLinearMap

lemma det_regularizedGradient_at_critical {f : ℂ → ℂ}
    (hf : DifferentiableOn ℂ f halfPlane) {k : ℝ} {ξ z : ℂ}
```

```
    (hz : z ∈ halfPlane) (hcrit : criticalMap f k z = ξ) :
    LinearMap.det (fderiv ℝ (regularizedGradient f k ξ) z).toLinearMap =
      -(k ^ 2 / z.im ^ 2) * ‖f z - ξ‖ ^ 2 *
        LinearMap.det (fderiv ℝ (criticalMap f k) z).toLinearMap
```

Key Consequence: The regularized gradient $\mathcal{G} = \texttt{regularizedGradient}(F, k, \xi)$ is smooth across the pole $F(z) = \xi$! Its zeros in $\mathbb{H}$ are disjoint and consist of:

(1) The **pole** $F(z) = \xi$ (at most one point in $\mathbb{H}$ by univalence of F), where $G_F(z) - \xi = \frac{iy}{k} F'(z) \neq 0$ and $\det_{\mathbb{R}} D\mathcal{G}(z) = k^2 y^{-2} |F'(z)|^4 \cdot \frac{y^2}{k^2 |F'(z)|^2} > 0$, so its Poincaré index is $+1$.
(2) The **local maxima** of V_ξ^F, where $G_F(z) = \xi$ and $J_F(z) < 0$, so $\det_{\mathbb{R}} D\mathcal{G}(z) > 0$ and the Poincaré index is $+1$.
(3) The **saddle points** of V_ξ^F, where $G_F(z) = \xi$ and $J_F(z) > 0$, so $\det_{\mathbb{R}} D\mathcal{G}(z) < 0$ and the Poincaré index is -1.

6.2. **Picard–Lindelöf Level Curves and Winding Interior (`LevelCurve.lean`, `LevelIndex.lean`).** For a regular level value $c > 0$ of u (meaning $\{u \leq c\}$ is compact in $\mathbb{H}$ and $\nabla u \neq 0$ on $\{u = c\}$), `LevelCurve.lean` defines the unit-speed-like tangent vector field

$$\mathcal{T}(z) := i \frac{\nabla u(z)}{\|\nabla u(z)\|^2} \qquad (\texttt{tangentGradient}).$$

Using Mathlib's Picard–Lindelöf theorem (`IsPicardLindelof`, `exists_isMIntegralCurve_of_isMIntegralCurve`), every point on the compact regular level set $\{u = c\}$ lies on a global periodic integral curve of $\mathcal{T}$ along which u is constant (`isConstant_along_flow`). After reparametrizing by period $T_i > 0$, $\{u = c\}$ decomposes into a finite disjoint union of 1-periodic C^1 simple closed curves $\gamma_1, \ldots, \gamma_m$ (`compact_regular_level_oriented_curves`) satisfying $\gamma_j'(t) = T_j \mathcal{T}(\gamma_j(t)) = i\alpha_j(t) \nabla u(\gamma_j(t))$ with $\alpha_j(t) > 0$.

- **Bypass of the Jordan Curve Theorem (`LevelIndex.lean`)**:

```
lemma sum_loopIndex_sublevel {f : ℂ → ℂ} (hf : DifferentiableOn ℂ f halfPlane)
    {k : ℝ} (hk : 0 < k) {ξ : ℂ} {c : ℝ} (hc : 0 < c)
    (hK : IsCompact {z | z ∈ halfPlane ∧ squaredReciprocalPotential f k ξ z ≤ c})
    (hreg : ∀ z ∈ halfPlane, squaredReciprocalPotential f k ξ z = c →
      regularizedGradient f k ξ z ≠ 0)
    ... (z : ℂ) (hz : z ∉ ⋃ i, range (γ i)) :
    (∑ i, loopIndex ⟨fun t : unitInterval => γ i t, (hγ i).1.continuous.comp
    continuous_subtype_val⟩ z) =
      if z ∈ {w | w ∈ halfPlane ∧ squaredReciprocalPotential f k ξ w < c} then 1 else 0
```

Proof Mechanism:

(1) First, on the open bounded set $\texttt{windingInterior}(\gamma_i) = \{z \notin \operatorname{range}\gamma_i : \operatorname{Ind}(\gamma_i, z) \neq 0\}$, whose frontier is contained in $\operatorname{range}\gamma_i \subset \{u = c\}$ (`frontier_windingInterior_subset`), the function $u(z) = |F(z) - \xi|^2 y^{-2k}$ cannot attain a local maximum at any point $z \in \mathbb{H}$ because $\log u(z) = 2\log|F(z) - \xi| - 2k \log y$ is harmonic where $F(z) \neq \xi$ except for $-2k \log y$, which has strictly positive Laplacian $\Delta(-2k \log y) = 2ky^{-2} > 0$ (`no_local_max_squaredReciprocalPotential`)! Thus $u < c$ on $\texttt{windingInterior}(\gamma_i)$ (`reciprocalPotential_lt_on_windingInterior`).
(2) Second, at any point $\gamma_i(0)$ on the curve, moving in the inward normal direction $i\gamma_i'(0) = -T_i \alpha_i(0) \nabla u(\gamma_i(0))$ decreases u below c (entering $\{u < c\}$) while increasing $\operatorname{Ind}(\gamma_i, \cdot)$ by $+1$ (`winding_jump_centred` from `Winding.lean`).
(3) Therefore, the integer-valued difference $\mathcal{J}(z) := \sum_{i=1}^m \operatorname{Ind}(\gamma_i, z) - \mathbf{1}_{\{u(z)<c\}}$ has jump $1 - 1 = 0$ across every curve γ_i, so $\mathcal{J}$ extends continuously to all of $\mathbb{C} \to \mathbb{Z}$ (`sum_loopIndex_eq_sublevel_i`). Since $\mathbb{C}$ is connected and $\mathcal{J}(z) = 0$ for $\|z\|$ large, $\mathcal{J} \equiv 0$ on $\mathbb{C}$.

6.3. **Equal Count of Maxima and Saddles (Lemma 5.5, `CriticalCount.lean`).**

Theorem 6.1 (`GoodPair.card_maxima_eq_card_saddles`, Lemma 5.5)**.** *Let (F, ξ) be a good pair for $k > 0$, and suppose ξ is in the image of G_F on $\{z \in \mathbb{H} : J_F(z) > 0\}$. Then F attains the value ξ on $\mathbb{H}$, and the finite sets of local maxima $\mathcal{M}_F(\xi) = \{z \in \mathbb{H} : G_F(z) = \xi,\ J_F(z) < 0\}$ and saddle points $\mathcal{S}_F(\xi) = \{z \in \mathbb{H} : G_F(z) = \xi,\ J_F(z) > 0\}$ are nonempty and have the same cardinality:*

$$\#\mathcal{M}_F(\xi) = \#\mathcal{S}_F(\xi).$$

Proof. Choose $c > 0$ larger than the value of u at all critical points of V_ξ^F, so that $\{u \le c\}$ is compact in $\mathbb{H}$ and contains $\mathcal{M}_F(\xi) \cup \mathcal{S}_F(\xi)$ in $\{u < c\}$, while $\{u = c\}$ is a regular level decomposed into oriented loops $\gamma_1, \dots, \gamma_m$. Along each γ_j, $\mathcal{G}(\gamma_j(t))$ is a positive scalar multiple of $-\nabla u(\gamma_j(t)) = i\gamma_j'(t)/\alpha_j(t)$, so by the Hopf tangent turning theorem (`tangent_turning_log`), $\mathrm{wind}(\mathcal{G} \circ \gamma_j) = +1$. On the other hand, applying the Poincaré–Hopf degree formula (`windingNumber_finite_regular_zeros`) to $\mathcal{G}$ on a convex box $K \subset \mathbb{H}$ containing $\{u \le c\}$ and summing over $j = 1, \dots, m$ gives

$$m = \sum_{j=1}^{m} \mathrm{wind}(\mathcal{G} \circ \gamma_j) = \sum_{a \in Z(\mathcal{G}) \cap K} \texttt{planeIndex}(D\mathcal{G}(a)) \sum_{j=1}^{m} \mathrm{Ind}(\gamma_j, a).$$

By `sum_loopIndex_sublevel`, $\sum_{j=1}^{m} \mathrm{Ind}(\gamma_j, a) = \mathbf{1}_{\{u(a) < c\}}$. Summing instead the tangent turning numbers $\sum_{j=1}^{m} \mathrm{Ind}(\gamma_j, a_0) = 1$ around a basepoint $a_0 \in \{u < c\}$ shows $\sum_{j=1}^{m} \mathrm{wind}(\mathcal{G} \circ \gamma_j) = 1$ (`sum_regularizedGradient_winding_eq_one`). Meanwhile, the zeros of $\mathcal{G}$ in $\{u < c\}$ are the pole $\{z : F(z) = \xi\}$ (index $+1$, cardinality 1 since u attains its minimum 0 inside $\{u < c\}$), the maxima $\mathcal{M}_F(\xi)$ (each of index $+1$), and the saddles $\mathcal{S}_F(\xi)$ (each of index -1):

$$1 = 1 + \#\mathcal{M}_F(\xi) - \#\mathcal{S}_F(\xi) \implies \#\mathcal{M}_F(\xi) = \#\mathcal{S}_F(\xi). \qquad \square$$

7. Equal-Rank Borel Pairing and Expected Area Transport (RankPairing–ExpectedTransport)

7.1. **Lexicographic Order and Equal-Rank Pairing (`RankPairing.lean`).**

To pair each saddle $z \in \mathcal{S}_F(\xi)$ bijectively with a local maximum $w \in \mathcal{M}_F(\xi)$ at the same critical value $\xi = G_F(z) = G_F(w)$ with $\mathrm{Im}\, w \ge \mathrm{Im}\, z$, `RankPairing.lean` orders critical points by the height quantity

$$\texttt{criticalHeight}(k, z) := \frac{y^k}{\|z - G_F(z)\|_*} = y^{k-1} \qquad \text{(so that } V_\xi^F(z) = k\,|F'(z)|^{-1} y^{k-1} \text{ when } G_F(z) = \xi).$$

Specifically, `criticalOrderTuple` embeds $\mathbb{C} \hookrightarrow \mathbb{R} \times_{\mathrm{Lex}} (\mathbb{R} \times_{\mathrm{Lex}} \mathbb{R})$ via $z \mapsto (-y^{k-1}, \mathrm{Re}\, z, \mathrm{Im}\, z)$, defining a strict total order `criticalPointLe` on $\mathbb{C}$ that is preserved by affine maps $\alpha_t(z) = x_0 + t_y z$ (`criticalPointLe_affine`).

- **Order Rank and Equal-Rank Pairing Map** ($f : \mathbb{C} \to \mathbb{C}$, $k \in \mathbb{R}$, $z \in \mathbb{H}$):

```
def orderRank (S : Set α) (x : α) : ℕ := (S ∩ Iio x).ncard

def criticalPair (f : ℂ → ℂ) (k : ℝ) (z : ℂ) : ℂ :=
  nthElement (maximaFiber f k (criticalMap f k z))
    (orderRank (saddleFiber f k (criticalMap f k z)) z)
```

  By Lemma 5.5 applied on every superlevel $\{V_\xi^F \ge \Lambda\}$ (`card_high_saddles_le_high_maxima`), for every threshold $\Lambda > 0$, the number of saddles in $\mathcal{S}_F(\xi)$ with $V_\xi^F \ge \Lambda$ is at most the number of maxima in $\mathcal{M}_F(\xi)$ with $V_\xi^F \ge \Lambda$. Because $|F'(z)| = \frac{k}{y}|F(z) - \xi|$ at any critical point $G_F(z) = \xi$, ordering by V_ξ^F on a fiber reduces to comparing heights, and the r-th

highest saddle $z \in \mathcal{S}_F(\xi)$ is paired with the r-th highest maximum $w = R_F(z) \in \mathcal{M}_F(\xi)$, which therefore satisfies $V_\xi^F(w) \geq V_\xi^F(z)$, i.e.,

$$\frac{(\operatorname{Im} w)^{k-1}}{|F'(w)|} \geq \frac{(\operatorname{Im} z)^{k-1}}{|F'(z)|} \iff (\operatorname{Im} w)^{4(k-1)}|F'(w)|^{-4} \geq (\operatorname{Im} z)^{4(k-1)}|F'(z)|^{-4},$$

and is affine-equivariant: $R_{F_t}(z) = \alpha_t^{-1}(R_F(\alpha_t(z)))$ (`class_criticalPair_reroot`).

7.2. Borel Branches and Joint Measurability (`BorelPairing.lean`).

- **Countable Borel Branches (`BorelBranches`) and Proposition 5.6 (`measurable_criticalPair`)**:

```
structure BorelBranches (k : ℝ) (s : ℝ) where
  dom : ℕ → Set (DiskClass × ℂ)
  inv : ℕ → DiskClass → ℂ → ℂ
  measurable_dom : ∀ n, MeasurableSet (dom n)
  pairwise_disjoint : Pairwise (Disjoint on dom)
  cover : ∀ p : DiskClass × ℂ, p.2 ∈ halfPlane → 0 < s * normalizedJacobian (classFun p
    .1) k p.2 →
    ∃ n, p ∈ dom n
  ...
```

Bypass of Lusin–Novikov Selection: At every point $(g_0, z_0) \in \mathcal{K} \times \mathbb{H}$ where $s \cdot J_{F_0}(z_0) > 0$ ($s \in \{+1, -1\}$), the map $(g, z) \mapsto (g, G_{\texttt{classFun}(g)}(z))$ has invertible partial derivative along $\mathbb{C}$, so by the inverse function theorem on Banach spaces (`HasStrictFDerivAt.toOpenPartialHomeomorph`) and continuity on $\mathcal{K}$, there is an open product neighborhood $U \times V \subset \mathcal{K} \times \mathbb{H}$ on which $z \mapsto G_F(z)$ is injective for every $F \in U$. Since $\mathcal{K} \times \mathbb{H}$ is second-countable and Lindelöf, a countable subcover disjointified into Borel sets $D_n = \texttt{dom}(n)$ (`countable_joint_injective_partition`) partitions $\{s \cdot J_F(z) > 0\}$. Because $\mathcal{K} \times \mathbb{H}$ is a Polish space, the injective continuous map $(g, z) \mapsto (g, G_F(z))$ on each D_n is a Borel measurable embedding (`ContinuousOn.measurableEmbedding`), yielding Borel partial inverses $\psi_n : \mathcal{K} \times \mathbb{C} \to \mathbb{H}$ (`exists_borel_partial_inverse`). Expressing the finite fibers $\mathcal{S}_F(\xi)$ and $\mathcal{M}_F(\xi)$ and their order ranks in terms of the countable branches $\psi_n(g, \xi)$ proves immediately that `criticalPair` is jointly Borel measurable (`measurable_criticalPair`).

7.3. Area Transport and the Contradiction $\rho \leq 2$ (`AreaTransport.lean`–`ExpectedTransport.lean`).

(1) **Fiber Sum Change of Variables (`AreaTransport.lean`, `transport_area`)**: Decomposing $\Omega_F^{\pm} = \{z \in \mathbb{H} : \pm J_F(z) > 0\}$ into the countable disjoint Borel branches D_n and applying Mathlib's injective change-of-variables formula on each branch proves that for any measurable indicator $u : \mathbb{C} \to [0, \infty]$,

$$\int_{\Omega_F^{\pm}} \underbrace{y^{\beta-1}|F'(z)|^{-2}}_{\texttt{transportDensity}} (\pm J_F(z))\, u(z)\, dA(z) = \int_{\mathbb{C}} \sum_{z \in \Omega_F^{\pm} \cap G_F^{-1}(\xi)} \underbrace{y^{\beta-1}|F'(z)|^{-4}}_{\texttt{targetDensity}} u(z)\, dA(\xi).$$

(2) **Pointwise Domination under Equal-Rank Pairing (`targetDensity_le_pair`)**: When $\beta - 1 = 4(k-1)$ (i.e., $k = (\beta+3)/4$), the potential inequality $V_\xi^F(w) \geq V_\xi^F(z)$ for $w = R_F(z)$ is *precisely* the target density inequality

$$(\operatorname{Im} z)^{\beta-1}|F'(z)|^{-4} \leq (\operatorname{Im} w)^{\beta-1}|F'(w)|^{-4}!$$

(3) **Box Exhaustion and Expected Transport Inequality (`TransportBoxes.lean`, `ExpectedTransport.l`**
Restricting z to the displacement window $W_N = \{1/N \leq \frac{\operatorname{Im} w}{\operatorname{Im} z} \leq N,\ |\operatorname{Re} w - \operatorname{Re} z| \leq N \operatorname{Im} z\}$ and the box $B_{R,Y} = [-R, R] \times [1, Y]$ ensures that $w = R_F(z)$ lands inside the enlarged box

$B'_{R,Y,N} = [-(R+NY), R+NY] \times [1/N, NY]$ (`pair_mem_enlargedBox`). Summing over fibers gives the pointwise box transport inequality (`transport_box`):

$$\int_{W_N(F)\cap B_{R,Y}} y^{\beta-1}|F'(z)|^{-2}J_F^+(z)\,dA(z) \le \int_{B'_{R,Y,N}} y^{\beta-1}|F'(w)|^{-2}J_F^-(w)\,dA(w).$$

Taking $\mathbb{E}_P[\,\cdot\,]$ using Tonelli's theorem and the scale-covariance identity (`weighted_averagedDensity` in `ExpectedDensity.lean`: $\mathbb{E}_P[y^{\beta-1}|H_F(z)|^2\Phi(F_z)] = y^{-1}\mathbb{E}_P[\Phi(F)]$) evaluates both box integrals explicitly in terms of $\int_{-R}^{R} dx \int_1^Y \frac{dy}{y} = 2R\log Y$:

$$2R(\log Y)\,\mathbb{E}_P\big[J_F^+(i)\,\mathbf{1}_{W_N}(F,i)\big] \le 2(R+NY)(\log Y + 2\log N)\,\mathbb{E}_P\big[J_F^-(i)\big].$$

Dividing by $2R\log Y$, letting $R\to\infty$, then $Y\to\infty$, and finally $N\to\infty$ by monotone convergence ($W_N \uparrow$ good pairs) yields Proposition 6.1 (`expected_positive_le_negative` and `expected_normalizedJacobian_nonpos`):

$$\mathbb{E}_P[J_F^+(i)] \le \mathbb{E}_P[J_F^-(i)] \implies \mathbb{E}_P[J_F(i)] \le 0.$$

(4) **Proposition 6.2: $\rho \le 2$ (`transferRho_le_two`, `transferBeta_le_one`)**: If $\rho > 2$, then $\beta = \log_2\rho > 1$. On the one hand, the Koebe/distortion bound gives $\beta \le 3$ (and even for general $\beta \in (1,3)$, $\mathbb{E}_P[J_F(i)] = \frac{(\beta-1)(3-\beta)}{16k^2} > 0$, while for $\beta \ge 3$ a smaller exponent choice still yields a contradiction; in `ExpectedTransport.lean`, `no_weighted_law_between_one_and_three` rules out $\beta \in (1,3)$ and `transferBeta_le_one` concludes $\beta \le 1$, i.e., $\rho \le 2$).

8. Proof of $B(-2) = 1$, Brennan's Conjecture, and Endpoint Sharpness (Spectrum–InverseMoment)

8.1. The Sharp Inverse-Square Spectrum $B(-2) = 1$ (`Koebe.lean`, `Spectrum.lean`).

- **Upper Bound $\beta_f(-2) \le 1$ (`beta_inverse_le_one`)**: Since $\rho \le 2$, for every $\varepsilon > 0$, Proposition 2.4 (`uniform_inverse_square_bound`) gives $M_{-2}(f,r) \le C_\varepsilon(1-r)^{-1-\varepsilon}$ for all $f \in \mathcal{S}$ and $r \in [1/2,1)$. Taking logarithms and dividing by $\log\frac{1}{1-r}$ as $r\to 1^-$ gives $\beta_f(-2) \le 1+\varepsilon$ for all $\varepsilon > 0$, hence $\beta_f(-2) \le 1$.
- **Lower Bound $\beta_{f_K}(-2) \ge 1$ (`koebe_beta_ge_one`)**: For the Koebe function $f_K(z) = \frac{z}{(1-z)^2} \in \mathcal{S}$ (`koebeMap_schlicht`), its derivative is $f_K'(z) = \frac{1+z}{(1-z)^3}$. On the boundary arc $-re^{i\theta}$ with $|\theta| \le 1-r$ and $r \in [1/2,1)$, $|f_K'(-re^{i\theta})| \le 2(1-r)$ (`koebe_arc_deriv_bound`), so $|f_K'(-re^{i\theta})|^{-2} \ge \frac{1}{4(1-r)^2}$. Integrating over $\theta \in [-(1-r), 1-r]$ gives $M_{-2}(f_K,r) \ge \frac{1}{4\pi(1-r)}$ (`koebe_mean_lower`), hence $\beta_{f_K}(-2) \ge 1$ and $B(-2) = 1$ (`inverse_spectrum_eq_one`).

8.2. Area Integrability on $\mathbb{D}$ for $t \in (-2, 2/3)$ and Brennan's Conjecture (`NegativeArea`–`DomainArea`).

(1) **Negative Powers $t \in (-2,0)$ (`NegativeArea.lean`, `schlicht_negative_area`)**: For any $\alpha > 0$, integrating $M_{-2}(f,r) \le C_{\alpha/2}(1-r)^{-1-\alpha/2}$ in polar coordinates against $(1-r)^\alpha r\,dr$ shows that $\int_{\mathbb{D}}(1-\|z\|)^\alpha|f'(z)|^{-2}\,dA(z) < \infty$ (`weighted_inverse_integrable`). For $t \in (-2,0)$, let $q = -t/2 \in (0,1)$ and $\alpha = \frac{1-q}{2q} > 0$, so that

$$|f'(z)|^t = \Big((1-\|z\|)^\alpha|f'(z)|^{-2}\Big)^q\Big((1-\|z\|)^{-1/2}\Big)^{1-q}.$$

Applying the weighted AM–GM inequality (`integrableOn_geometric_product`) and $\int_{\mathbb{D}}(1-\|z\|)^{-1/2}\,dA < \infty$ yields $\int_{\mathbb{D}}|f'(z)|^t\,dA(z) < \infty$.

(2) **Smoothed Modulus Means and Positive Powers $t \in (0, 2/3)$ (`SmoothMeans.lean`, `PositiveArea.lean`)**: To avoid singularities of $|f(z)|^{2q}$ at $f(z) = 0$, `SmoothMeans.lean` introduces the C^∞ bracket power

$$\langle w\rangle^2 := 1 + \|w\|^2 \quad (\texttt{bracketSq}), \qquad \psi_q(w) := (1+\|w\|^2)^q \quad (\texttt{smoothPower}),$$

which satisfies $0 \le \Delta\psi_q(w) = 4q(1+\|w\|^2)^{q-2}(1+q\|w\|^2) \le 4q(1+\|w\|^2)^{q-1}$ (`laplacian_smoothPower_boun`
Pulling back to the logarithmic strip $z = x + iy \mapsto e^{iz} = e^{-y}e^{ix} \in \mathbb{D}$ (`logDisk`), where $y \le 3(1 + \|f(e^{iz})\|^2)^{-1/4}$ by Koebe growth (`logStrip_height_bound`), the periodic rectangular trace theorem (`periodic_conformal_trace_bounded`) bounds

$$\sup_{0<r<1} \int_{-\pi}^{\pi} (1 + \|f(re^{i\theta})\|^2)^q \, d\theta \le C_q < \infty \qquad \forall q \in (0, 1/4) \quad (\texttt{schlicht_smooth_means}).$$

Combining this with $(1+\|f(z)\|^2)^{a-q} \lesssim (1-\|z\|)^{-4(a-q)}$ proves $\int_{\mathbb{D}}(1+\|f(z)\|^2)^a \, dA(z) < \infty$ for all $a < 1/2$ (`schlicht_function_area` in `PositiveArea.lean`). Meanwhile, for every $\eta > 0$, changing variables $w = f(z)$ via `conformal_lintegral_le` gives $\int_{\mathbb{D}} |f'(z)|^2(1 + \|f(z)\|^2)^{-1-\eta} \, dA(z) \le \int_{\mathbb{C}}(1+\|w\|^2)^{-1-\eta} \, dA(w) < \infty$ (`schlicht_weighted_derivative_integrable`). For $t \in (0, 2/3)$, setting $q = t/2 \in (0, 1/3)$, $\eta = \frac{1-3q}{4q} > 0$, and $a = \frac{(1+\eta)q}{1-q} = \frac{1+q}{4(1-q)} < \frac{1}{2}$, the geometric decomposition

$$|f'(z)|^t = \Big(|f'(z)|^2(1 + \|f(z)\|^2)^{-1-\eta}\Big)^q \Big((1 + \|f(z)\|^2)^a\Big)^{1-q}$$

implies $\int_{\mathbb{D}} |f'(z)|^t \, dA(z) < \infty$ (`schlicht_positive_area`).

(3) **Brennan's Conjecture (`DomainArea.lean`, `conformal_domain_area`)**: For any conformal bijection $\varphi : W \to \mathbb{D}$ with inverse $f = \varphi^{-1} : \mathbb{D} \to W$, normalizing $f_0(z) = (f(z) - f(0))/f'(0) \in \mathcal{S}$ shows $\int_{\mathbb{D}} |f'(w)|^t \, dA(w) < \infty$ for all $t \in (-2, 2/3)$ (`univalent_disk_area`). For $p \in (4/3, 4)$, we have $t := 2 - p \in (-2, 2/3)$, and the conformal change of variables $z = f(w)$ (`conformal_lintegral_eq`) yields

$$\int_W |\varphi'(z)|^p \, dA(z) = \int_{\mathbb{D}} |f'(w)|^2 \, |\varphi'(f(w))|^p \, dA(w) = \int_{\mathbb{D}} |f'(w)|^{2-p} \, dA(w) < \infty.$$

8.3. **Endpoint Divergence for the Koebe Function (`AffineArea`–`InverseMoment`).** Finally, Modules 41–47 prove that the open intervals $t \in (-2, 2/3)$ and $p \in (4/3, 4)$ are sharp:

- Near $z = -1$ ($z = -1 + w$ with $w = re^{i\theta}$, $0 < r < 1/2$, $|\theta| < \pi/3$), $-1 + w \in \mathbb{D}$ (`polar_left_mem_disk`) and $|f_K'(-1 + w)| = \frac{\|w\|}{\|2-w\|^3} \le \|w\| = r$, so $|f_K'(-1 + w)|^{-2} \ge r^{-2}$ (`koebe_polar_left_lower`).
- Near $z = +1$ ($z = 1 - w$ with $w = re^{i\theta}$, $0 < r < 1/2$, $|\theta| < \pi/3$), $1 - w \in \mathbb{D}$ (`polar_right_mem_disk`) and $|f_K'(1 - w)| = \frac{\|2-w\|}{\|w\|^3} \ge r^{-3}$, so $|f_K'(1 - w)|^{2/3} \ge r^{-2}$ (`koebe_polar_right_lower`).
- By `Sector.lintegral_eq_top_of_sector_lower_bound` in `SectorDivergence.lean`, integrating $r^{-2} \cdot r \, dr \, d\theta = r^{-1} \, dr \, d\theta$ over $(0, 1/2) \times (-\pi/3, \pi/3)$ diverges to $+\infty$. By monotone interpolation on $\mathbb{R}$, $\int_{\mathbb{D}} |f_K'(z)|^t \, dA(z) = \infty$ for all $t \notin (-2, 2/3)$, and via $p = 2 - t$ (`koebe_inverse_areaMoment`), $\int_{W_K} |\varphi_K'(w)|^p \, dA(w) = \infty$ for all $p \notin (4/3, 4)$.